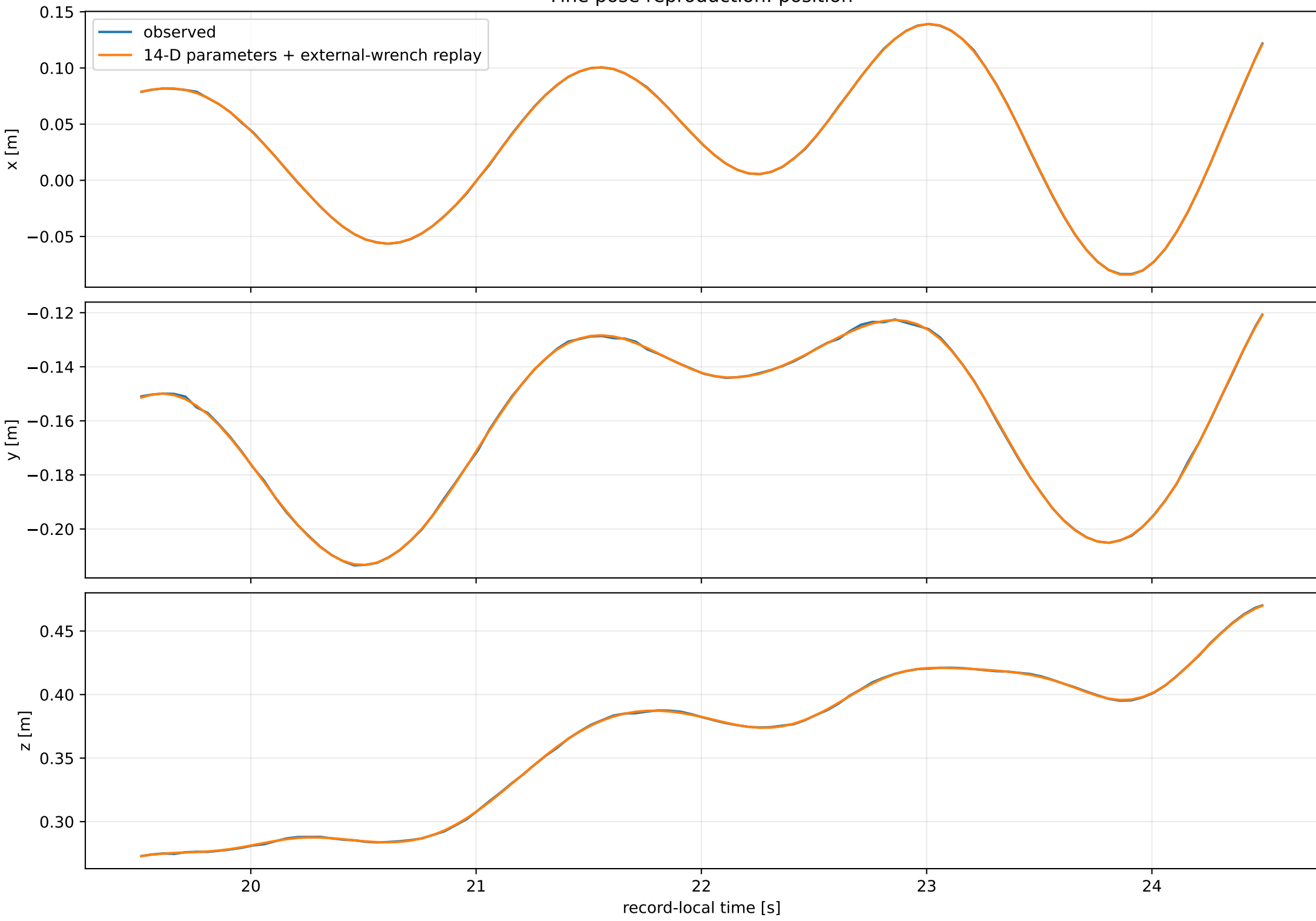
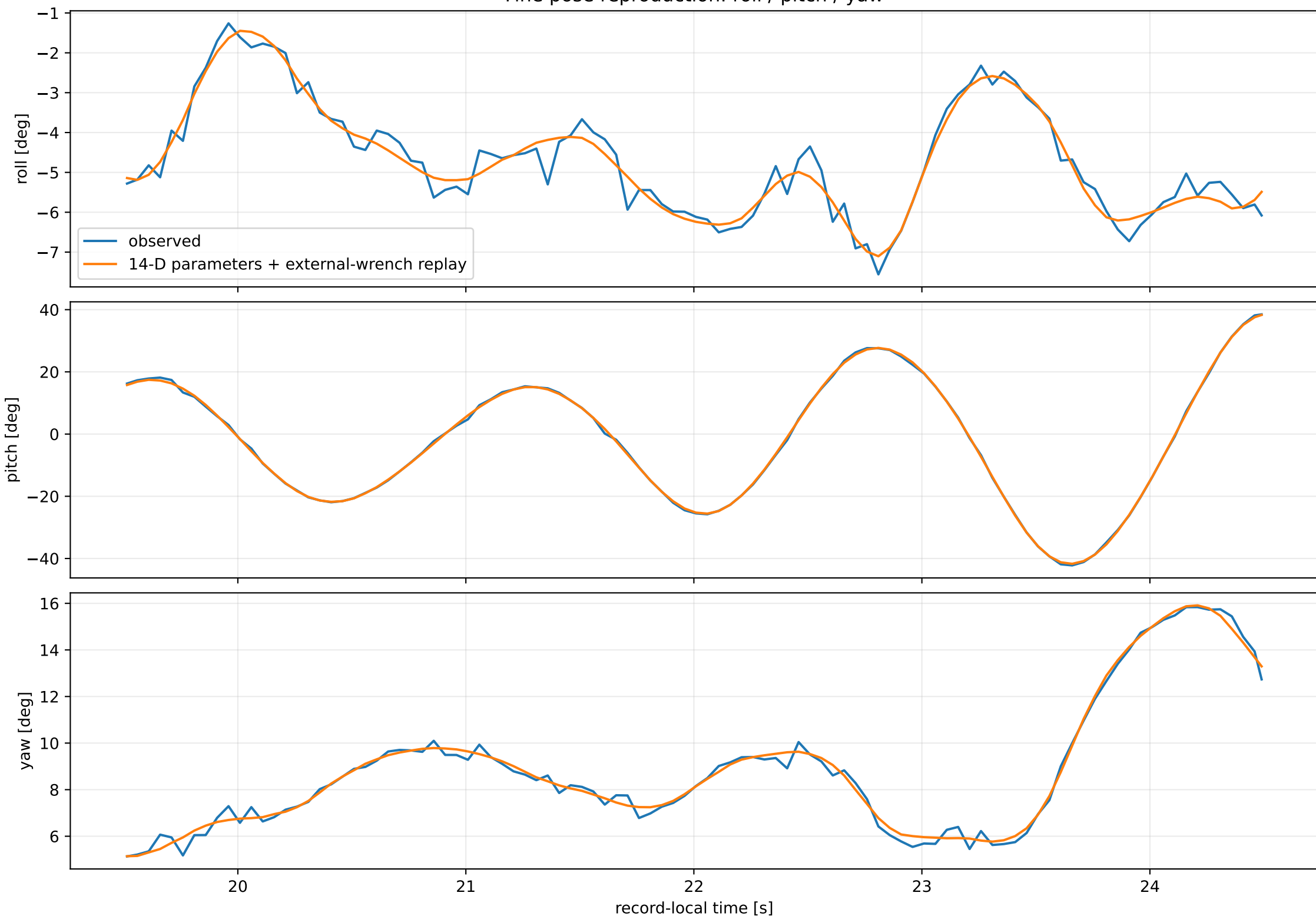


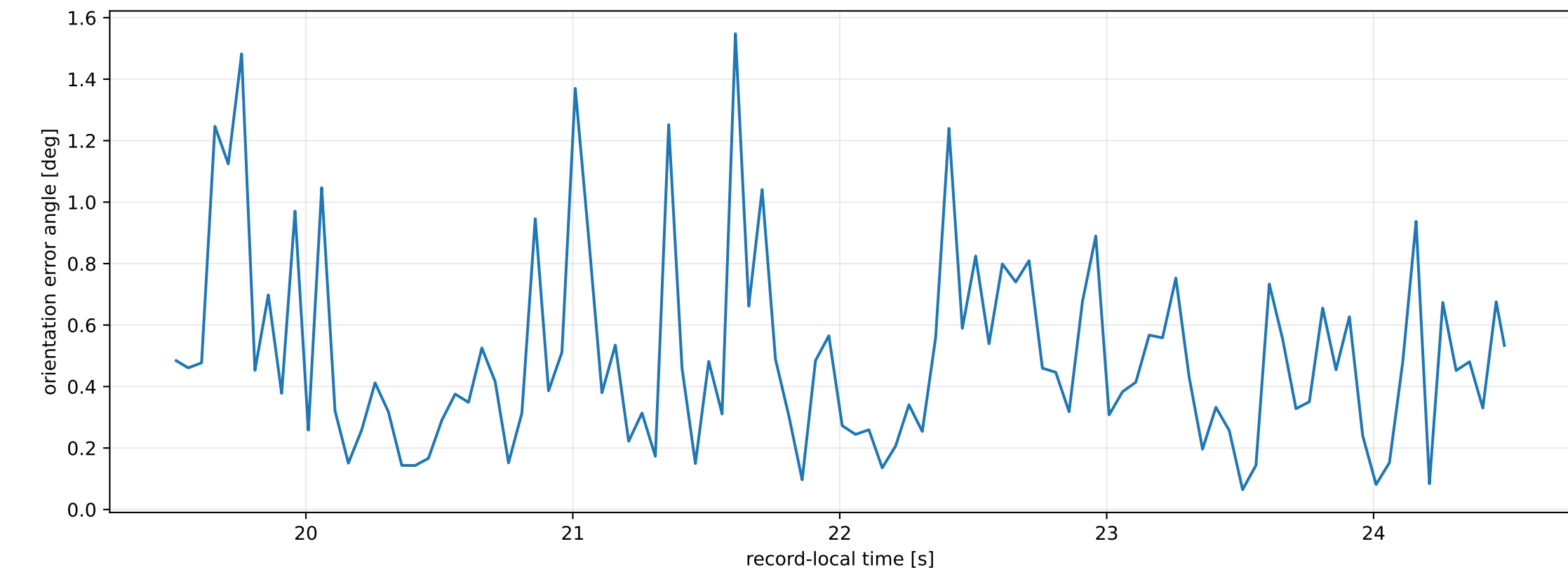
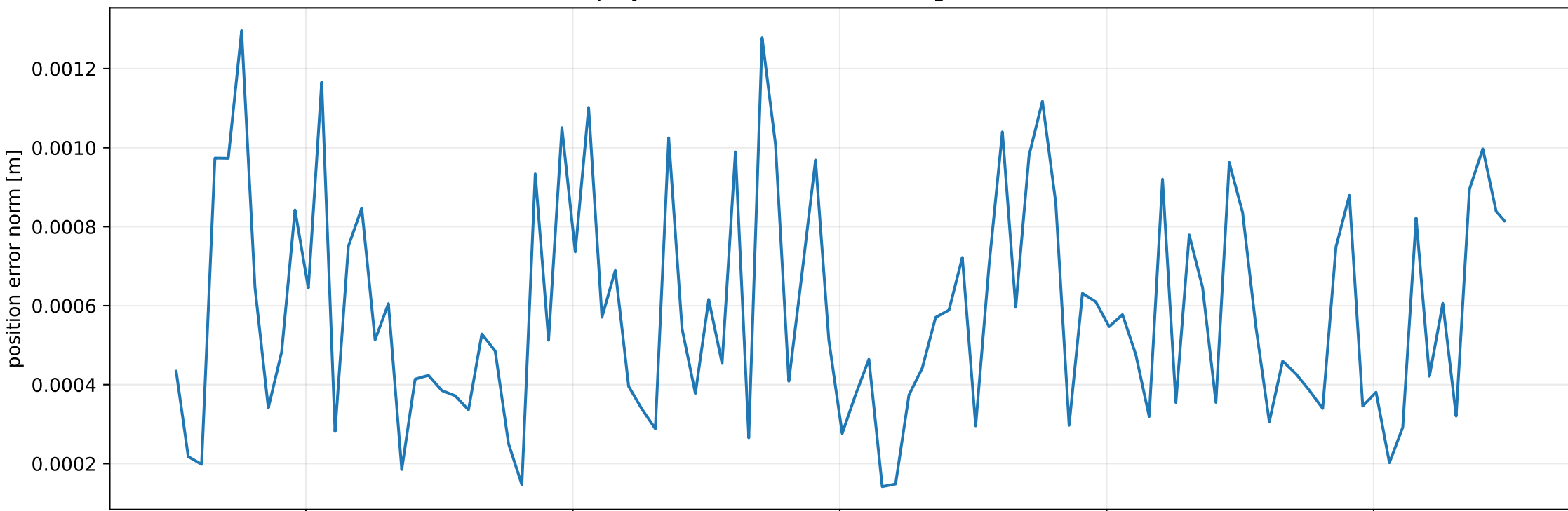
Fine pose reproduction: position



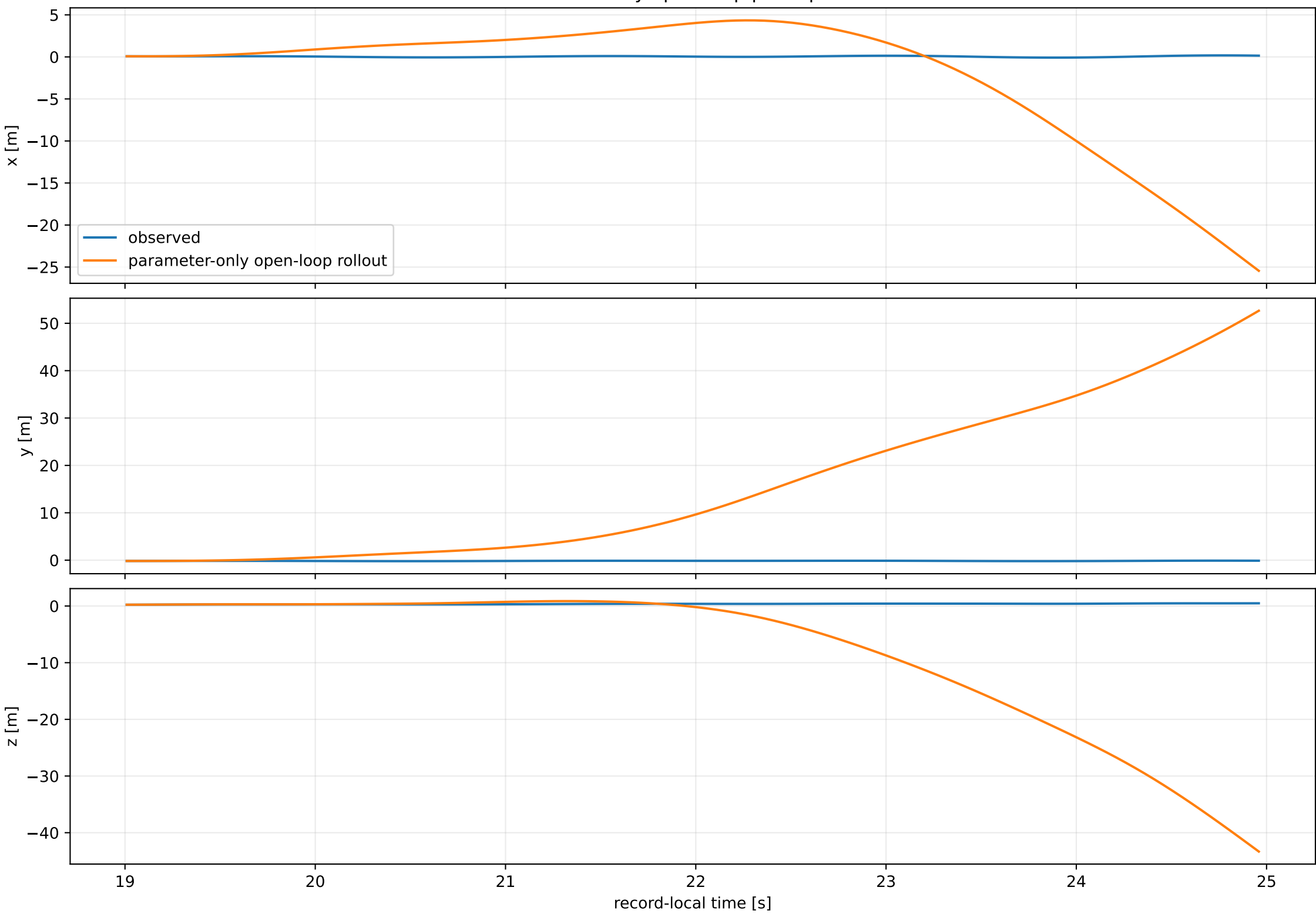
Fine pose reproduction: roll / pitch / yaw



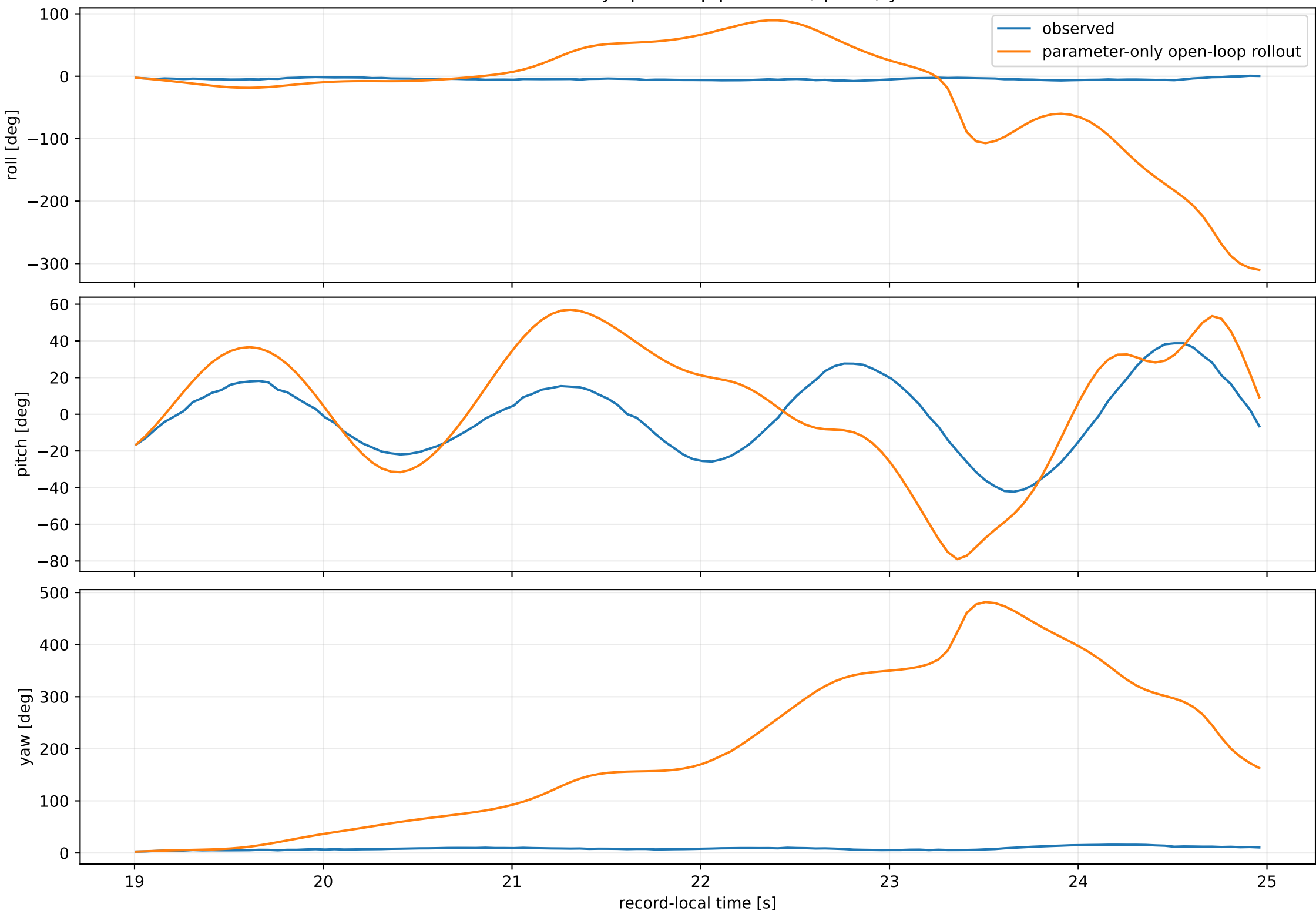
Fine replay error after the 14-D change — lower is better



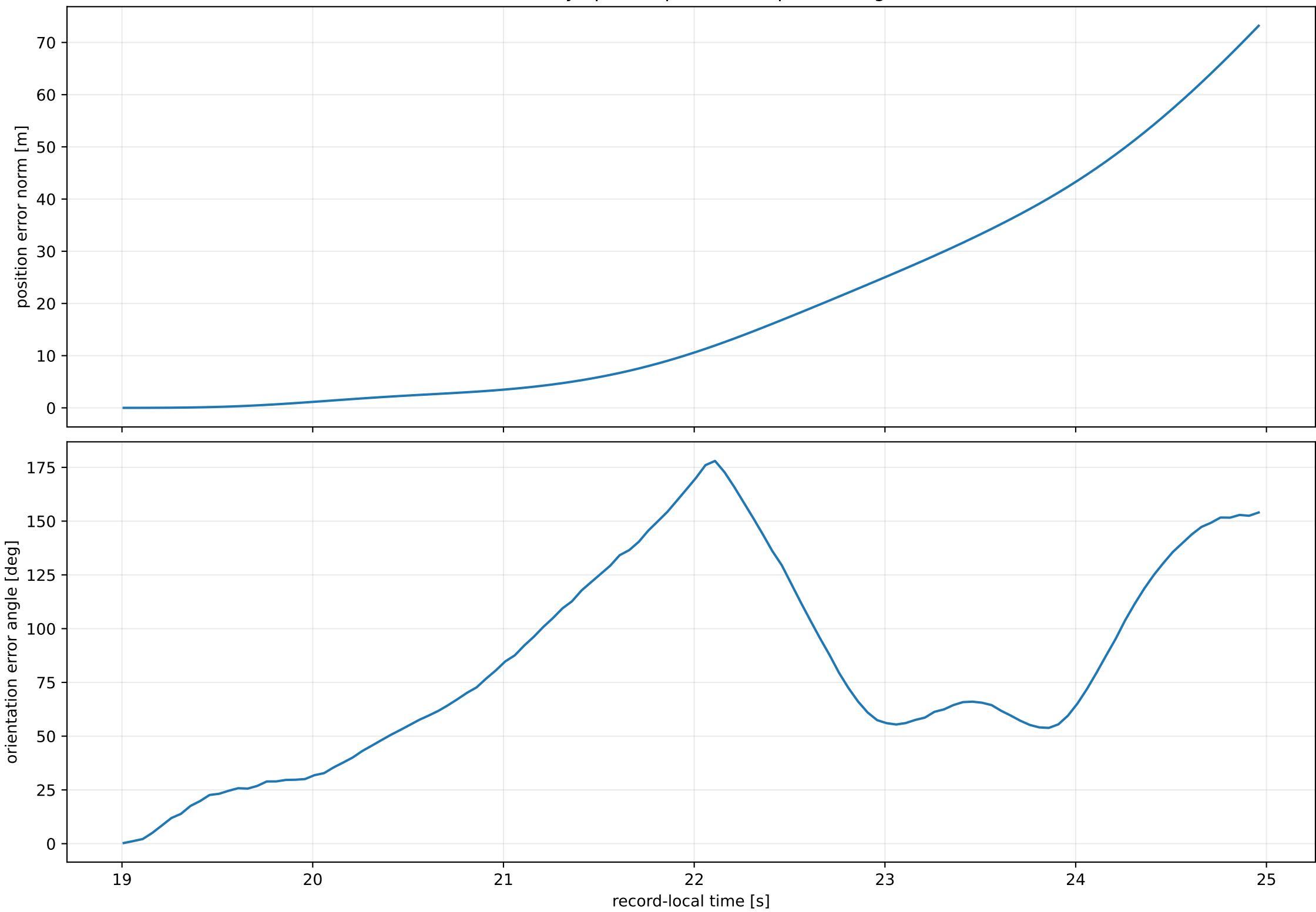
Parameter-only open-loop pose: position



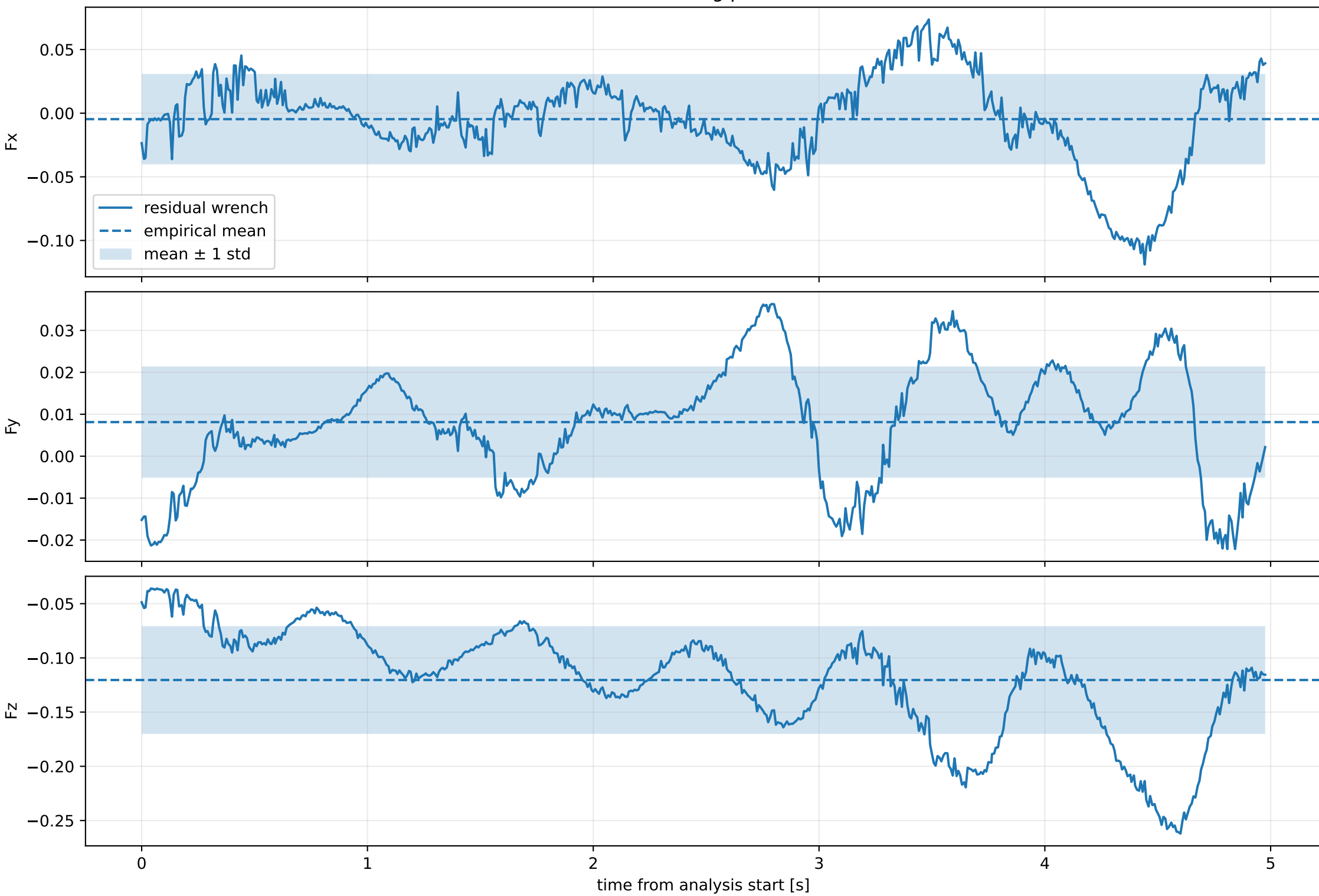
Parameter-only open-loop pose: roll / pitch / yaw



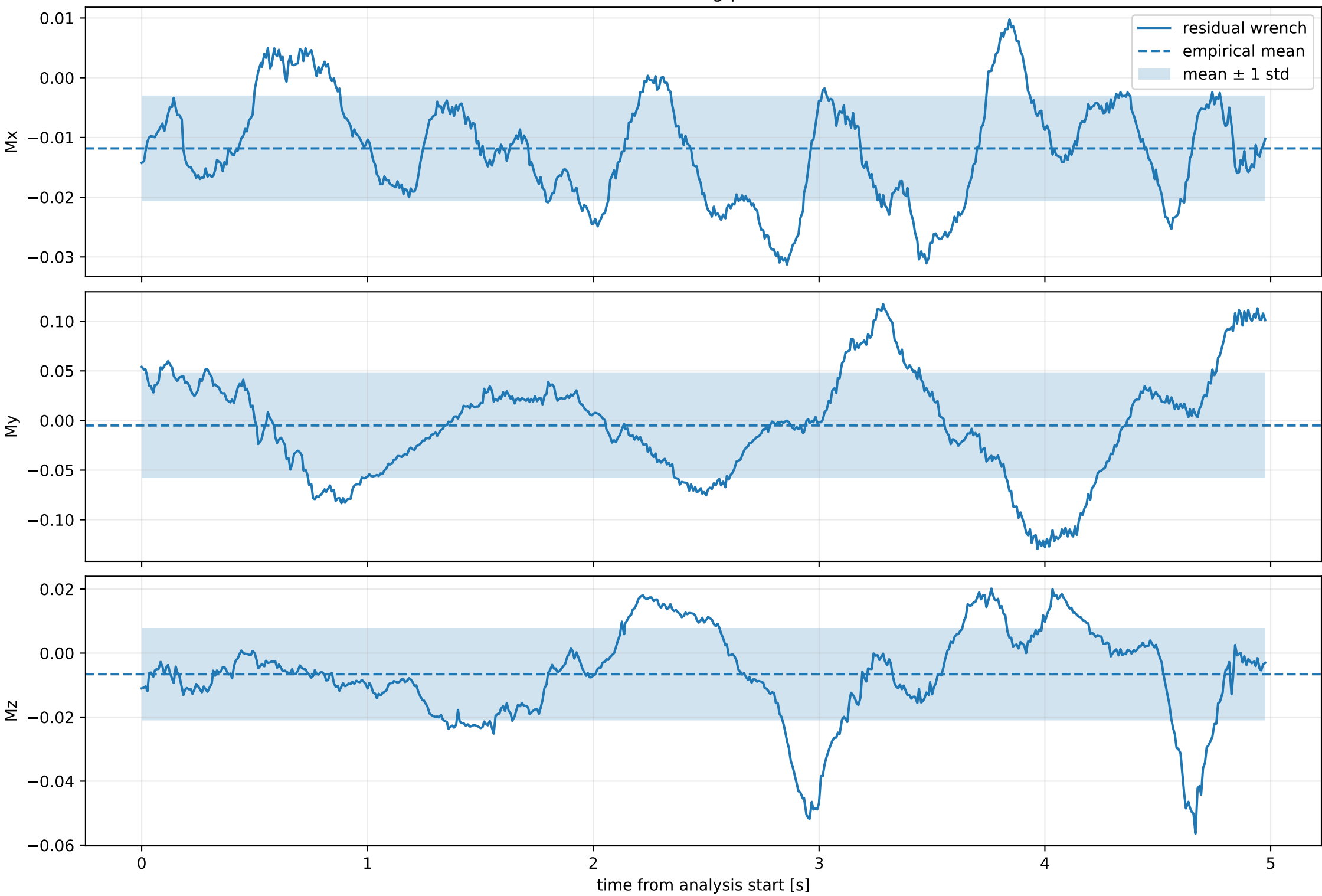
Parameter-only open-loop rollout (separate diagnostic)



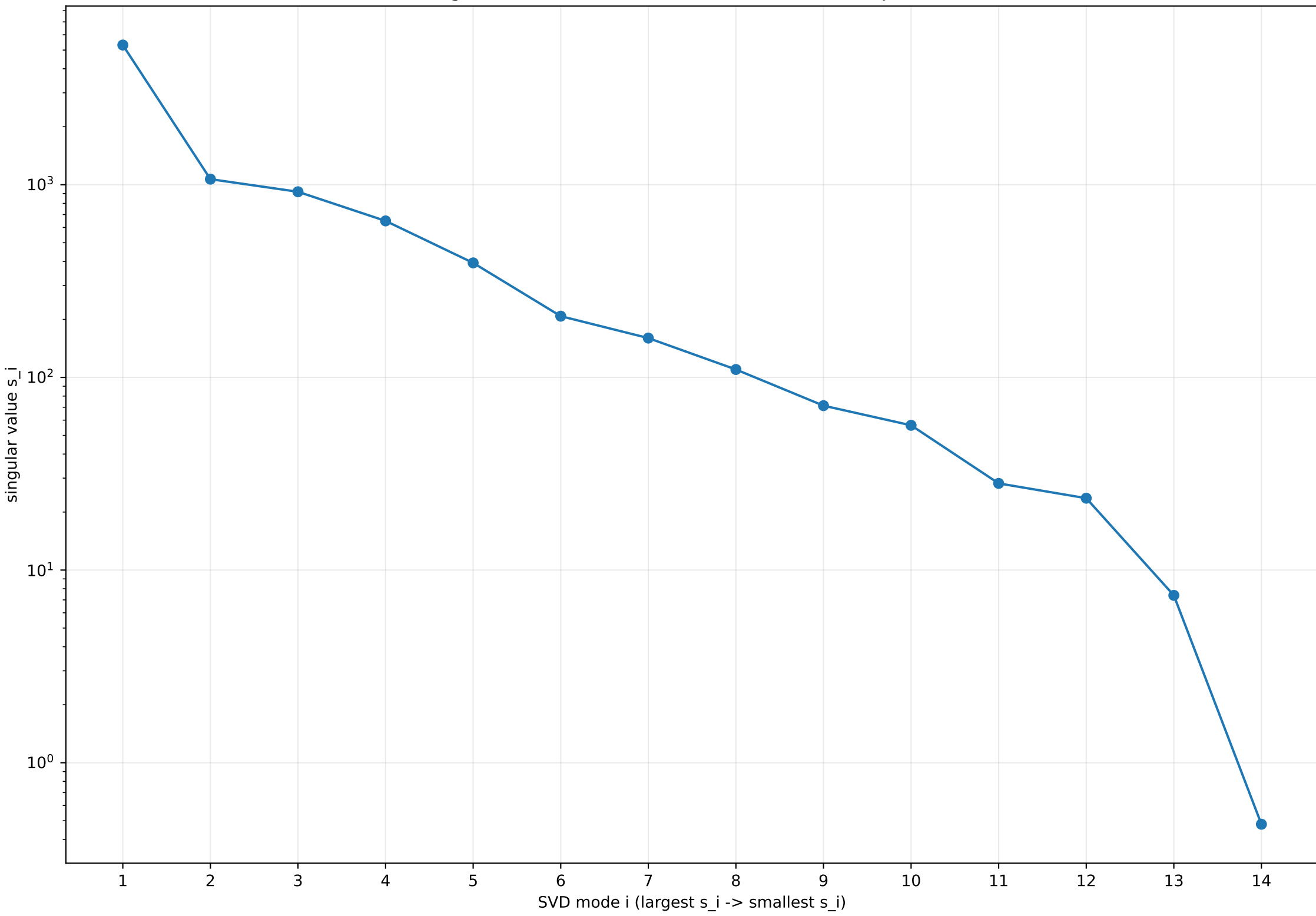
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



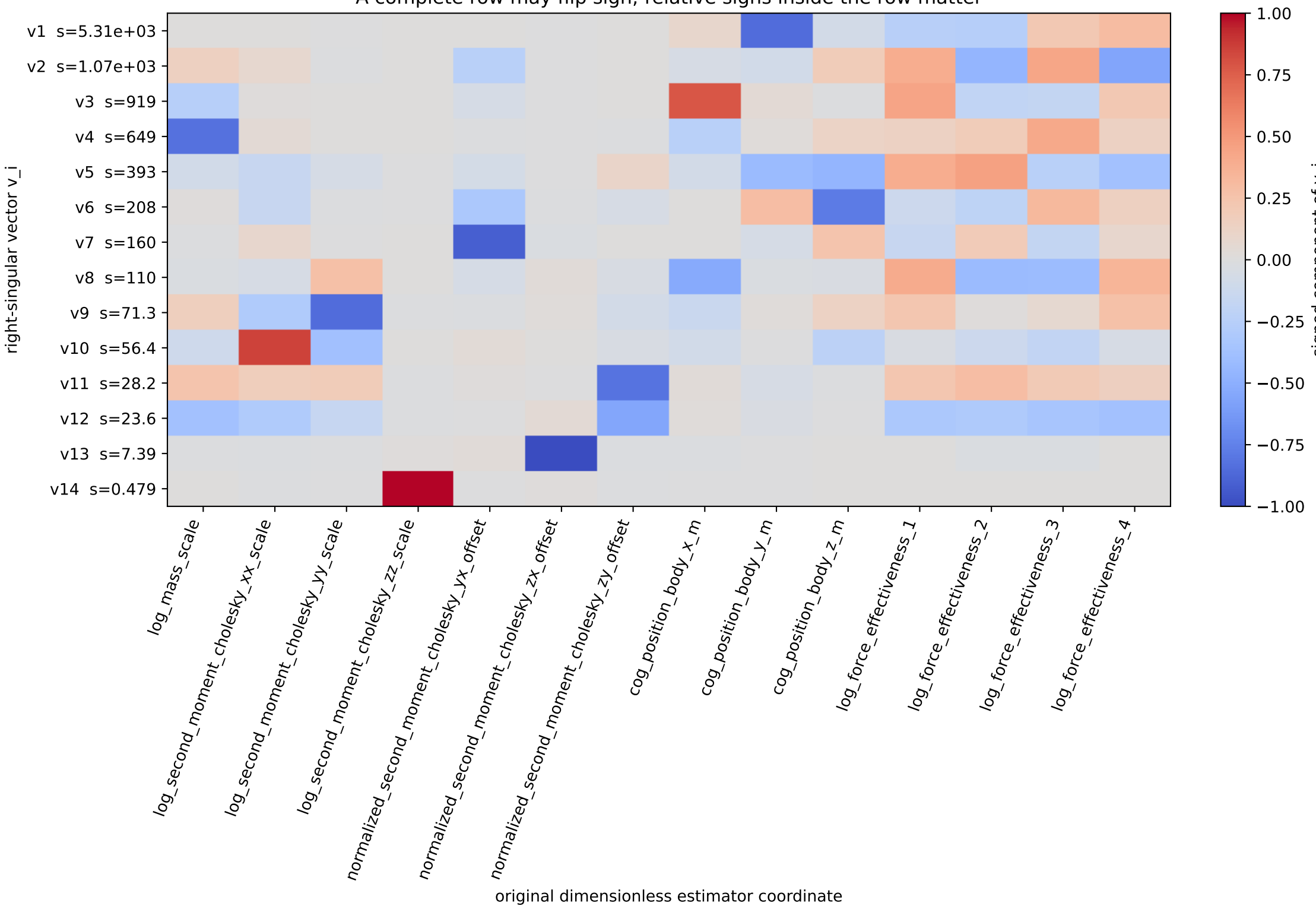
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36579803383

inertia J [kg m²]

[0.065011259 -2.5090274e-05 2.2580027e-05]
[-2.5090274e-05 0.06525007 9.368878e-06]
[2.2580027e-05 9.368878e-06 0.12899418]

principal moments [kg m²]

[0.065008643, 0.065252677, 0.12899419]

CoG position in body frame [m]

[-0.0023671012, 0.00087384913, 0.010520707]

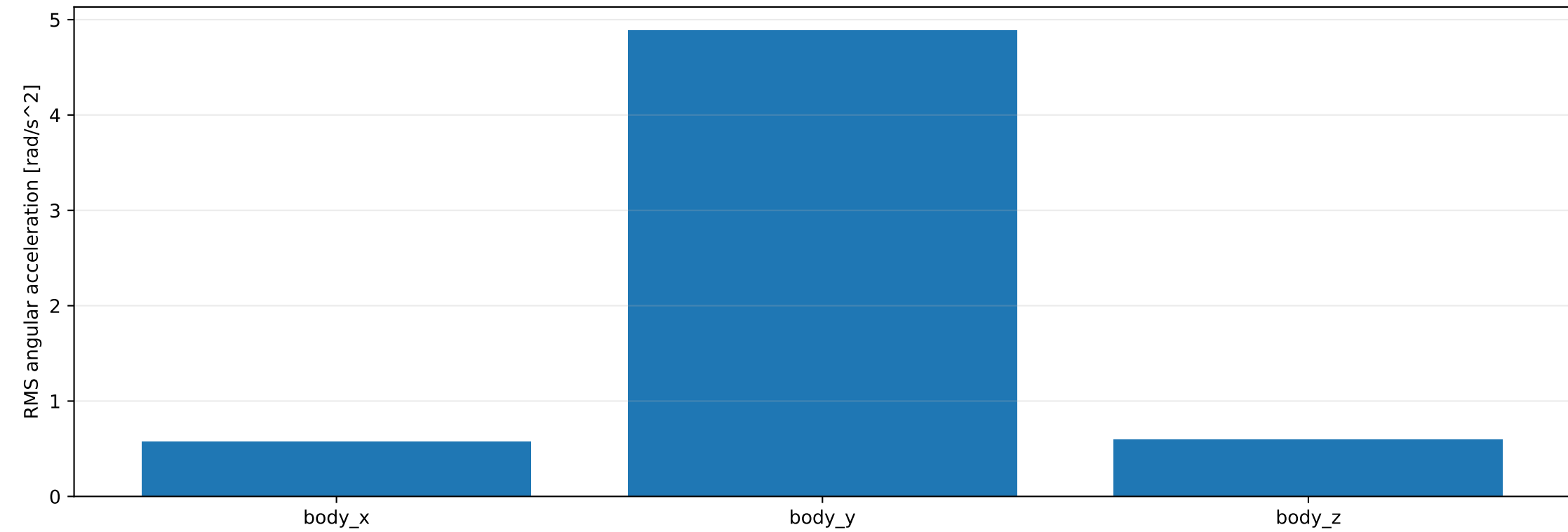
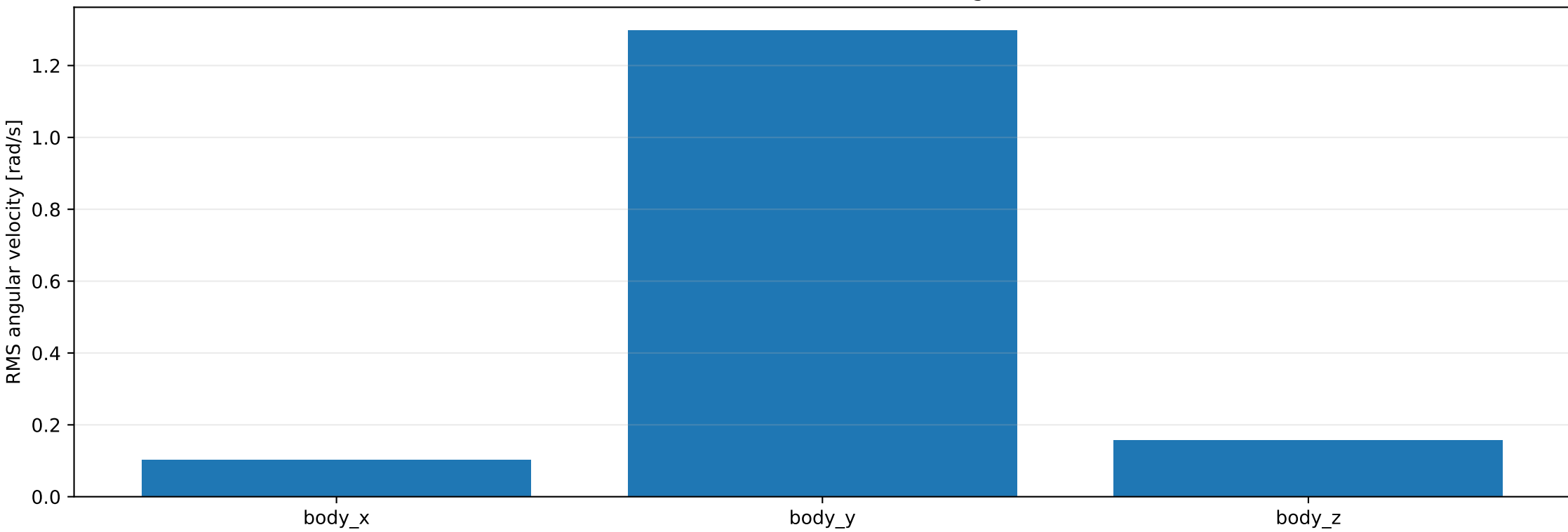
rotor force effectiveness

[0.96536116, 0.97320888, 0.9582561, 0.95494088]

selected delay [s]

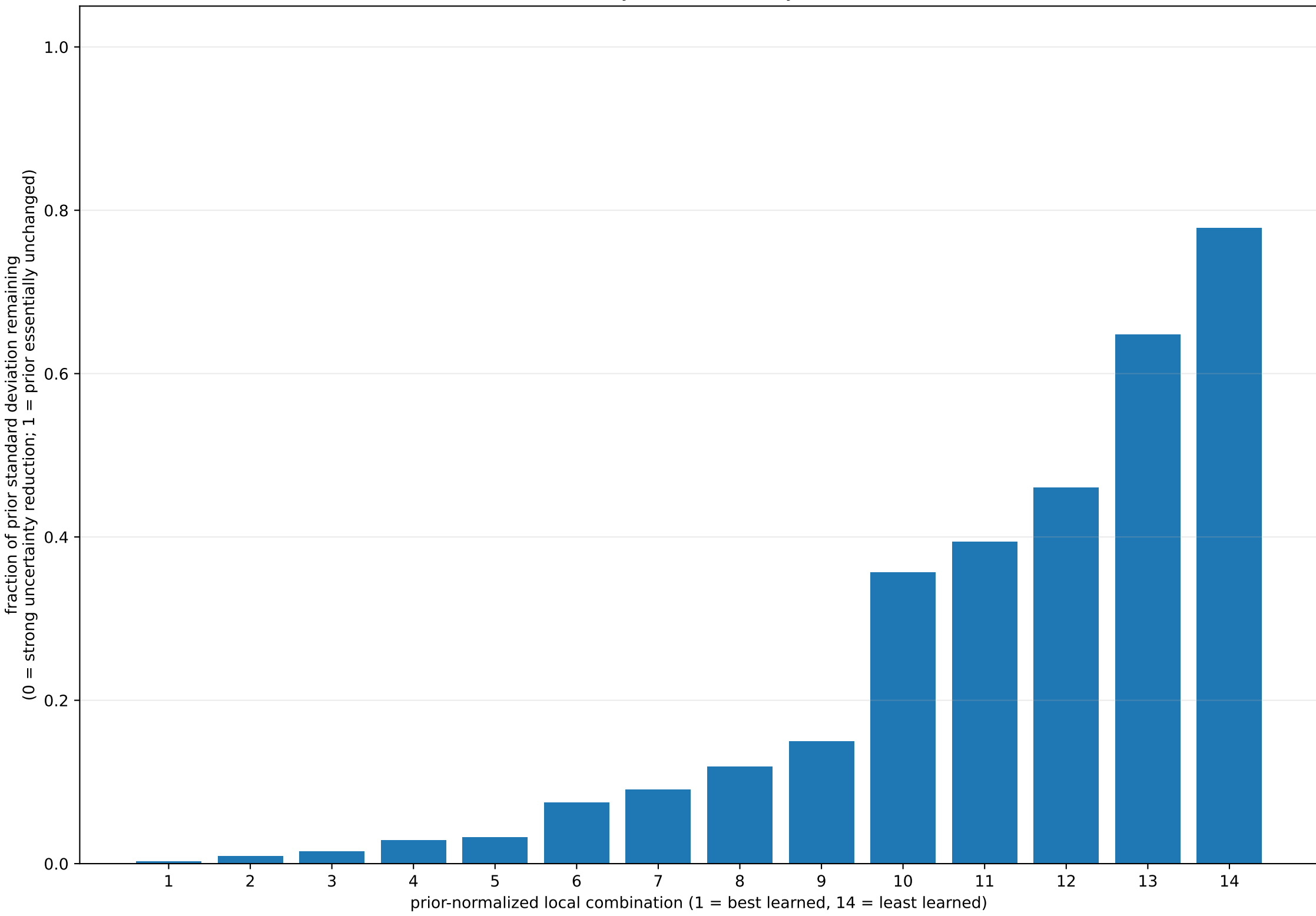
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

